Ultimate Pro Email Agent

Gmail AI Auto Reply System (Python + Ollama + Google API)

Overview

An intelligent **Gmail automation agent** that reads emails, understands context using AI, and generates **human-like replies automatically**.

It runs completely **locally using Ollama**, ensuring **privacy + control + zero cloud dependency**.

Key Features

Smart Gmail inbox reader
Filters spam, OTP, promotions, newsletters
Detects urgent emails automatically
AI-powered reply generation (LLM-based)
Smart modes:

- Student replies
- HR / Job / Internship replies
- Faculty / Academic replies
- Friendly personal replies

in Hindi + English (Hinglish support)

Fully automated OR manual approval mode
Fully local AI (no data sent to cloud)
Auto mark emails as read

🧠 Tech Stack

Technology	Purpose
🐍 Python	Core backend logic
📧 Gmail API	Email access
🔐 OAuth 2.0	Authentication
🧠 Ollama LLM	AI reply generation
🌐 Requests	API communication

Project Structure

```
gmail-agent/
│
├── gmail_agent.py      # Main automation script
└──credentials.json     # Google OAuth credentials
```

Setup Guide

1. Clone Repository

```
git clone
```

https://github.com/yourusernameKKname/gmail-agent.git

---

2. Install Dependencies

```
pip install google-api-python-client google-auth-httplib2
pip install google-auth-oauthlib requests
```

---

3. Google Cloud Setup

1. Open Google Cloud Console
2. Create Project
3. Enable **Gmail API**
4. Configure OAuth Consent Screen
5. Create **OAuth Client ID (Desktop App)**
6. Download credentials.json
7. Place it in project folder

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4. Install Ollama AI

Download:
👉 https://ollama.com/download

Run model:

```
ollama run phi3
```

Other models:

```
ollama run llama3
ollama run mistral
```

Run Project

python gmail_agent.py

✓ First run opens browser login
✓ Google authentication required
✓ Token saved automatically

Configuration

Edit inside script:

```
AUTO_SEND = True     # Auto send replies
MODEL = "phi3"       # AI model
CHECK_INTERVAL = 20  # Email check time (seconds)
```

How It Works

Step 1: Email Fetch

- Reads unread Gmail emails
- Removes spam / OTP / ads

Step 2: AI Understanding

- Detects intent:
  - Job / Internship
  - Academic / Faculty
  - Personal / Friends
- Detects urgency level

Step 3: AI Reply

- Sends prompt to Ollama
- Generates human-like reply
- Adjusts tone automatically

Step 4: Send Response

- Auto send OR manual approval
- Marks email as read

Sample Output

📩 NEW EMAIL
⚡ PRIORITY: URGENT
👤 FROM: HR Company
📌 SUBJECT: Interview Scheduled

🧠 AI RESPONSE:

Thank you for scheduling the interview.
I confirm my availability and look forward to it.

Regards,
Rohit Singh

Privacy Advantage

🔒 100% local AI processing
🚫 No external AI APIs
📪 Emails never leave your system
🧠 Fully offline LLM support

Use Cases

- 🎓 Students managing internships/job emails
- 💼 Professionals handling HR communication
- 👷 Freelancers managing clients
- ⚡ Inbox automation
- 🤖 AI + automation portfolio project

Future Improvements

- 🌐 Web dashboard (Flask / React)
- 📊 Email analytics dashboard
- 🧠 Fine-tuned personal AI model
- 📱 WhatsApp / Telegram integration
- 📬 Smart priority inbox sorting

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